

BINOMIAL THEOREM

1.1 BINOMIAL EXPRESSION :

An algebraic expression consisting of two different terms is called a binomial expression.

e.g. (1) $x + y$ (2) $x^3 + y^3$

But $(x + nx)$ is not a binomial, it is called a monomial.

1.2 BINOMIAL THEOREM :

In elementary algebra, the binomial theorem describes the algebraic expansion of powers of a binomial expression. According to the theorem it is possible to expand the powers $(x + y)^n$ into a sum involving terms of the form ax^by^c , where exponents b and c are non-negative integers with $b + c = n$ and the coefficient 'a' of each term is a specific positive integer depending on n and b .

This theorem was given by Newton.

Binomial Theorem $\Rightarrow (x + y)^n$ $\begin{cases} \text{If } n \in \mathbb{N} \\ \text{(Form a finite series)} \\ \text{Any index } n \notin \mathbb{N} \end{cases}$

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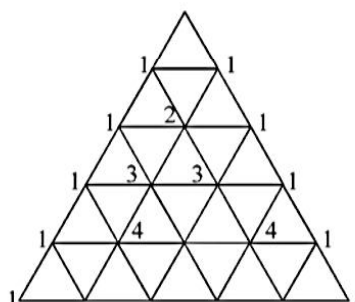
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Binomial Theorem $\Rightarrow (x + y)^n$ $\begin{cases} \text{If } n \in \mathbb{N} \\ \text{(Form a finite series)} \\ \text{Any index } n \notin \mathbb{N} \\ \text{(Form an infinite series)} \end{cases}$

1.3 HISTORICAL DEVELOPMENT :

Earlier people used to multiply the brackets to expand the given binomial of known index.

Then came the Pascal triangle



Note that

- The powers of x go down until it reaches zero, starting value is n .
- The power of y goes up from zero until it reaches n .
- The n^{th} row of the Pascals triangle will be the coefficients of the expanded binomial.

1.4 STATEMENT OF THE THEOREM :

$$(x + y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_r x^{n-r} y^r + \dots + {}^nC_n x^{n-n} y^n.$$

$$\text{We observe } T_1 = {}^nC_0 x^n$$

$$T_2 = {}^nC_1 x^{n-1} \cdot y^1$$

$\Rightarrow$ General term in the expansion of $(x + y)^n$ is

$$T_{r+1} = {}^nC_r x^{n-r} \cdot y^r$$

Where nC_r is called as combinatorial or binomial coefficient also denoted by $\binom{n}{r}$.

$$\text{Also, } (x + y)^n = \sum_{r=0}^n {}^nC_r x^{n-r} \cdot y^r$$

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Proof-1:

Binomial theorem for a positive integral index is a special case of a very general result called symmetric product which states that

$$(x + y_1)(x + y_2) \dots (x + y_n) = x^n + S_1 x^{n-1} + S_2 x^{n-2} + S_3 x^{n-3} + \dots + S_n.$$

where S_r ($r = 1, 2, 3, \dots, n$) denotes sum of product of quantities $y_1, y_2, \dots, y_n$ taken r at a time.

$$\text{Thus, } S_1 = y_1 + y_2 + \dots + y_n \text{ (} {}^nC_1 \text{ terms)}$$

$$S_2 = y_1 y_2 + y_2 y_3 + \dots \text{ (} {}^nC_2 \text{ terms)}$$

$$S_3 = y_1 y_2 y_3 + y_2 y_3 y_4 + \dots \text{ (} {}^nC_3 \text{ terms)}$$

and so on

Now put

$$y_1 = y_2 = y_3 = \dots = y_n = y$$

We get

$$S_1 = y + y + y + \dots + y \text{ (} {}^nC_1 \text{ terms)} = {}^nC_1 y$$

$$S_2 = y^2 + y^2 + \dots + y^2 \text{ (} {}^nC_2 \text{ terms)} = {}^nC_2 y^2$$

and so on

In this special case

$$(x + y)^n = x^n + {}^nC_1 x^{n-1} \cdot y + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_n y^n.$$

Proof-2: Combinatorial Proof

The coefficient of xy^2 in $(x+y)^3 = (x+y)(x+y)(x+y)$.

$$= x x x + x x y + x y x + x y y + y x x + y x y + y y x + y y y$$

$$= x^3 + 3x^2y + 3xy^2 + y^3$$

Equals ${}^3C_2 = 3$ because there are three x, y strings of length 3 with exactly two y 's namely $x y y, y x y, y y x$.

Corresponding to the three 2 element subsets of $\{1, 2, 3\}$ namely $\{2, 3\}, \{1, 3\}, \{1, 2\}$ where each subset specifies the position of y in corresponding string.

Similarly in $(x+y)^n$

1.5 IMPORTANT POINTS OF EXPANSION :

- (1) Number of terms in expansion of $(x+y)^n$ is $n+1$ i.e., one more than index.

or

By begger's method n coins and 2 beggars

$$\therefore {}^{n+1}C_1 \Rightarrow (n+1) \text{ times.}$$

Illustration :

$$\therefore {}^{n+1}C_1 \Rightarrow (n+1) \text{ times.}$$

Illustration :

Find number of term is the expansion of $(x+y+z+w)^{10}$.

Sol. Coin = 10, beggers = 4

${}^{13}C_3$ terms. **Ans.**

- (2) Sum of indices of x and y in each term in the expansion of $(x+y)^n$ is n .

- (3) New expansions by $(x+y)^n$

(a) We have $(x+y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1}y + {}^nC_2 x^{n-2}y^2 + \dots + {}^nC_n y^n$ (1)

i.e., $(x+y)^n = \sum_{r=0}^{r=n} {}^nC_r x^{n-r} \cdot y^r$.

- (b) Replace y by $-y$

$$(x-y)^n = {}^nC_0 x^n - {}^nC_1 x^{n-1}y + {}^nC_2 x^{n-2}y^2 + \dots + {}^nC_r x^{n-r} \cdot y^r (-1)^r + \dots + {}^nC_n (-1)^n \cdot y^n$$

i.e., $(x-y)^n = \sum_{r=0}^{r=n} {}^nC_r (-1)^r x^{n-r} \cdot y^r$.

(c) Now replace x by 1 and y by x in Ist

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_r x^r + \dots + {}^nC_n y^n.$$

Also, i.e., $(1+x)^n = \sum_{r=0}^n {}^nC_r x^r.$

(d) $(1-x)^n = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_r (-1)^r x^r + \dots + {}^nC_n (-1)^n x^n.$

i.e., $(1-x)^n = \sum_{r=0}^n (-1)^r {}^nC_r x^r.$

(e) Also remember

$$(1+x)^n + (1-x)^n = 2 [{}^nC_0 + {}^nC_2 x^2 + \dots]$$

$$\text{and } (1+x)^n - (1-x)^n = 2 [{}^nC_1 x + {}^nC_3 x^3 + {}^nC_5 x^5 + \dots]$$

$\therefore$ Coefficient of x^r in the expansion of $(1+x)^n$ is nC_r .

$$T_{r+1} = {}^nC_r x^r$$

$\therefore$ coefficient of $(r+1)^{\text{th}}$ term = coefficient of $x^r = {}^nC_r$ in the expansion of $(1+x)^n$.

For e.g., Find the coefficient of x^6 in $(1+3x+3x^2+x^3)^{15} = [(1+x)^3]^{15} = (1+x)^{45}$

$$\therefore T_{r+1} = {}^{45}C_r x^r$$

$$\therefore r = 6.$$

$$\therefore \text{coefficient is } {}^{45}C_6.$$

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Illustration :

Find the value of $(1+\sqrt{2})^7 + (1-\sqrt{2})^7 = ?$

Sol. $(1+\sqrt{2})^7 + (1-\sqrt{2})^7 = 2 \left[{}^7C_0 1 + {}^7C_2 (\sqrt{2})^2 + \dots + {}^7C_6 (\sqrt{2})^6 \right] = 476.$

Illustration :

Find the value of $(\sqrt{3}+3)^5 - (\sqrt{3}-3)^5 = ?$

Sol. $(\sqrt{3}+3)^5 - (\sqrt{3}-3)^5 = (\sqrt{3})^5 \left[(1+\sqrt{3})^5 - (1-\sqrt{3})^5 \right]$
 $= 2 \cdot 9 \cdot \sqrt{3} \left[{}^5C_1 (\sqrt{3}) + {}^5C_3 (\sqrt{3})^3 + {}^5C_5 (\sqrt{3})^5 \right] = 2376.$

Illustration :

Find sum of series upto n terms $\sum_{r=0}^n (-1)^r {}^nC_r \left[\left(\frac{1}{2}\right)^r + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \dots \text{upto } m \text{ terms} \right]$

Sol. $T_1 = \sum_{r=0}^n {}^nC_r (-1)^r \left(\frac{1}{2}\right)^r = \left(1 - \frac{1}{2}\right)^n = \frac{1}{2^n}$

$$T_2 = \sum_{r=0}^n \left(\frac{3}{4}\right)^r (-1)^r {}^nC_r = \left(1 - \frac{3}{4}\right)^n = \left(\frac{1}{4}\right)^n$$

$\vdots$

$T_1 + T_2 + T_3 + \dots \text{upto } m \text{ terms}$

$$\left(\frac{1}{2}\right)^n + \left(\frac{1}{2}\right)^{2n} + \left(\frac{1}{2}\right)^{3n} \dots \text{upto } m \text{ terms}$$

$$= \frac{1}{2^n} \left[\frac{1 - \left(\frac{1}{2^n}\right)^m}{1 - \frac{1}{2^n}} \right] = \frac{2^{mn} - 1}{2^n - 1}$$

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- (4) ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are called binomial coefficient or combinatorial coefficients and may be simply written as $C_0, C_1, C_2, \dots, C_n$.

$$(x + y)^n = {}^nC_0 x^n y^0 + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_n x^0 y^n.$$

Find the sum of all the combinatorial coefficient.

i.e., ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$.

Put $x = 1$ and $y = 1$ to get sum of all the binomial coefficient.

$(x + 2y)^n$ find the sum of all the coefficients.

$$(x + 2y)^2 = x^2 + 4xy + 4y^2$$

$\therefore$ Sum of all coefficient $= 1 + 4 + 4 = 9$.

We can also get it by putting $x = y = 1$

$$(1 + 2)^2 = 9$$

$\therefore$ Sum of all binomial coefficients in

$$(x + y)^n = 2^n \rightarrow \text{In this case sum of coefficient} = \text{sum of binomial coefficient.}$$

- (5) Binomial coefficients of the term equidistant from beginning and end are equal.

$$(2x + 3y)^2 = {}^2C_0 (2x)^2 + {}^2C_1 (2x)^1 (3y)^1 + {}^2C_2 (2x)^0 (3y)^2.$$

Now coefficient of 1st and last term is same ${}^2C_0 = {}^2C_2$.

(6) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

(7) $\left(\frac{n+1}{r+1}\right) {}^nC_r = {}^{n+1}C_{r+1}$

- (8) Consecutive binomial coefficient.

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} \text{ i.e., } \frac{{}^5C_4}{{}^5C_3} = \frac{5-4+1}{4} = \frac{1}{2}.$$

Illustration :

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If three consecutive coefficients in the expansion of $(1+x)^n$ be 165, 330 and 462. Find number of terms in the expansion of $(1+x)^n$.

Sol. $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}.$

Let three consecutive coefficients of terms are ${}^nC_{r-1}, {}^nC_r, {}^nC_{r+1}$

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} = \frac{330}{165} = 2. \quad \dots\dots(1)$$

$$\frac{{}^nC_{r+1}}{{}^nC_r} = \frac{n-(r+1)+1}{r+1} = \frac{n-r}{r+1}$$

$$\frac{n-r}{r+1} = \frac{462}{330} = \frac{n-r}{r+1} = \frac{7}{5}$$

$$n = 11 \text{ and } r = 4.$$

Practice Problem

- Q.1 If a, b, c, d are the coefficient of any four consecutive terms in the expansion of $(1+x)^n$, $n \in \mathbb{N}$ prove that $\frac{a}{a+b} + \frac{c}{c+d} = \frac{2b}{b+c}$.
- Q.2 $(x+\sqrt{2})^4 + (x-\sqrt{2})^4$.
- Q.3 If $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$.
Find (a) $a_0 + a_1 + a_2 + \dots + a_{2n}$
(b) $a_0 - a_1 + a_2 - a_3 + \dots + a_{2n}$.

Answer key

- Q.2 $2x^4 + 24x^2 + 8$ Q.3 (a) 3^n (b) 1

2.1 IMPORTANT TERMS IN BINOMIAL :

- (A) General term
(B) Term independent of x

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- (A) General term
(B) Term independent of x.
(C) Middle term

(A) GENERAL TERM :

$(T_{r+1})^{\text{th}}$ term is called as general term in $(x+y)^n$ and general term is given by

$$T_{r+1} = {}^nC_r x^{n-r} \cdot y^r$$

Illustration :

Find fourth term in the expansion of $\left(2x - \frac{y}{2}\right)^7$.

Sol. $T_{r+1} = {}^7C_r (2x)^{7-r} \cdot \left(-\frac{y}{2}\right)^r$

Put $r = 3$ to get 4th term

$$T_4 = -{}^7C_3 (2x)^4 \cdot \left(\frac{y}{2}\right)^3 = -70x^4 y^3.$$

Illustration :

Find term involving x^3 & x^4 in $\left(2x^2 - \frac{1}{3x}\right)^6$

Sol. Let T_{r+1} involving x^3 and x^4 then

$$\begin{aligned} T_{r+1} &= {}^6C_r (2x^2)^{6-r} \left(\frac{-1}{3x}\right)^r \\ &= {}^6C_r (2)^{6-r} \cdot \left(\frac{-1}{3}\right)^r x^{12-2r} \cdot \left(\frac{1}{x}\right)^r \\ &= {}^6C_r 2^{6-r} \left(\frac{-1}{3}\right)^r x^{12-3r} \end{aligned}$$

$$\Rightarrow \text{Coefficient of } x^3 \Rightarrow 12 - 3r = 3.$$

$$r = 3 \quad 4^{\text{th}} \text{ term involve } x^3 \text{ in expansion.}$$

$$\Rightarrow \text{Coeff. of } x^4 \Rightarrow 4 = 12 - 3r$$

$$r \neq \frac{8}{3}. \text{ So there is no term exist.}$$

Important : If r is non integral value then there is no term exist.

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Illustration :

Prove that coefficient of x^{50} in $(1+x^2)^{25} (1+x^{25}) (1+x^{40}) (1+x^{47})$ is equal to $1 + {}^{25}C_5$.

Sol. As we are interested in coefficient of x^{50} , we shall ignore all terms with exponent more than 50.

$$\begin{aligned} \text{So } (1+x^2)^{25} (1+x^{25}) (1+x^{40}) (1+x^{47}) \\ &= (1 + {}^{25}C_1 x^2 + \dots + {}^{25}C_{25} x^{50}) \times (1 + x^{25} + x^{40} + x^{45} + x^{47}) \\ &= {}^{25}C_{25} + {}^{25}C_5 = 1 + {}^{25}C_5. \end{aligned}$$

(B) TERM INDEPENDENT OF x :

It means term containing x^0 .

Illustration :

Find term independent of x in $\left(x^2 + \frac{1}{x^2} - 2\right)^{10}$.

$$\text{Sol. } \left(x^2 + \frac{1}{x^2} - 2\right)^{10} = \left(x - \frac{1}{x}\right)^{20}$$

$$\Rightarrow T_{r+1} = {}^{20}C_r x^{20-r} (-1)^r \frac{1}{x^r} = {}^{20}C_r x^{20-2r} (-1)^r$$

$$\Rightarrow 20 - 2r = 0 ; r = 10$$

$$\Rightarrow 11^{\text{th}} \text{ term is independent of } x.$$

(C) MIDDLE TERM :

Let T_m is middle term in expansion $(x + y)^n$ then

Case I : If n is odd, then number of terms will be even so there is two middle terms

$$\left(\frac{n+1}{2}\right)^{\text{th}} \text{ and } \left(\frac{n+3}{2}\right)^{\text{th}}.$$

Case II : If n is even, then number of terms will be odd so only one term is middle term $\left(\frac{n}{2} + 1\right)^{\text{th}}$.

Note : Binomial coefficient of middle term is greatest.

Illustration :

Find coefficient of t^8 in the expansion of $(1 + 2t^2 - t^3)^9$.

Sol. $T_{r+1} = {}^9C_r t^{2r} (2 - t)^r$

Now coefficient of t^8 appears in $r = 3, r = 4$.

$$r = 3 \Rightarrow {}^9C_3 t^6 (2 - t)^3 = {}^9C_3 \times t^6 \times {}^3C_p 2^{3-p} (-t)^p$$

$$\text{Put } p = 2 \Rightarrow {}^9C_3 \times {}^3C_2 \times 2^1 \times t^8 = 84 \times 6 \times t^8$$

Similarly when $r = 4$

$${}^9C_4 t^8 (2 - t)^4 = {}^9C_4 \times t^8 \times {}^4C_p 2^{4-p} (-t)^p$$

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$${}^9C_4 t^8 (2 - t)^4 = {}^9C_4 \times t^8 \times {}^4C_p 2^{4-p} (-t)^p$$

$$\text{Put } p = 0 \Rightarrow {}^9C_4 \times {}^4C_0 \times 2^4 \times t^8 = 126 \times 16 \times t^8$$

Hence coefficient of t^8 in the expansion of $(1 + 2t^2 - t^3)^9$ is $84 \times 6 + 126 \times 16$
 $= 504 + 2016 = 2520$. **Ans.**

Alternative:

We have

$$\begin{aligned} \left((1 + 2t^2) - t^3\right)^9 &= {}^9C_0 (1 + 2t^2)^9 - {}^9C_1 (1 + 2t^2)^8 \cdot t^3 + {}^9C_2 (1 + 2t^2)^7 \cdot t^6 \\ &\quad - {}^9C_3 (1 + 2t^2)^6 \cdot t^9 + \dots - {}^9C_9 (t^3)^9 \end{aligned}$$

$\therefore$ Coefficient of t^8 in the expansion of $(1 + 2t^2 - t^3)^9$

$$\begin{aligned} &= {}^9C_0 (\text{coefficient of } t^8 \text{ in } (1 + 2t^2)^9) - {}^9C_1 (\text{coefficient of } t^5 \text{ in } (1 + 2t^2)^8) \\ &\quad + {}^9C_2 (\text{coefficient of } t^2 \text{ in } (1 + 2t^2)^7) \end{aligned}$$

$$= {}^9C_0 \cdot {}^9C_4 2^4 - 0 + {}^9C_2 \cdot {}^7C_1 \cdot 2 = \frac{9 \cdot 8 \cdot 7 \cdot 6}{4 \cdot 3 \cdot 2 \cdot 1} \cdot 16 + \frac{9 \cdot 8}{2 \cdot 1} \cdot 7 \cdot 2$$

$$= 9 \cdot 8 \cdot 7 \cdot 5 = 2520. \text{ Ans.}$$

Illustration :

Prove that ${}^4C_4 \cdot {}^{100}C_6 + {}^4C_3 \cdot {}^{100}C_7 + {}^4C_2 \cdot {}^{100}C_8 + {}^4C_1 \cdot {}^{100}C_9 + {}^4C_0 \cdot {}^{100}C_{10} = {}^{104}C_{10}$

Sol.
$$\begin{aligned} & \underbrace{{}^{100}C_6 + {}^{100}C_7} + 3({}^{100}C_7 + {}^{100}C_8) + 3({}^{100}C_8 + {}^{100}C_9) + {}^{100}C_9 + {}^{100}C_{10} \\ &= {}^{101}C_7 + 3 {}^{101}C_8 + 3 {}^{101}C_9 + {}^{101}C_{10} \\ &= \underbrace{{}^{101}C_7 + {}^{101}C_8} + 2({}^{101}C_8 + {}^{101}C_9) + {}^{101}C_9 + {}^{100}C_{10} \\ &= {}^{102}C_8 + 2 \cdot {}^{102}C_9 + {}^{102}C_{10} = {}^{102}C_8 + {}^{102}C_9 + {}^{102}C_9 + {}^{102}C_{10} \\ &= {}^{103}C_9 + {}^{103}C_{10} = {}^{104}C_{10} \equiv {}^{104}C_{94} \end{aligned}$$

Alternative : Proof by PNC

$${}^4C_4 \cdot {}^{100}C_6 + {}^4C_3 \cdot {}^{100}C_7 + {}^4C_2 \cdot {}^{100}C_8 + {}^4C_1 \cdot {}^{100}C_9 + {}^4C_0 \cdot {}^{100}C_{10}$$

Out of 104 students of which 100 are boys and 4 are girls, we have to select 10 students.

This can be done in ${}^{104}C_{10} = {}^{104}C_{94}$

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Hence, $x + y = 114$ or 198 Ans.

Illustration :

Prove that middle term in the expansion of $(1+x)^{2n}$ is $\frac{1 \cdot 3 \cdot 5 \dots (2n-1) \cdot 2^n}{n!} \cdot x^n$.

Sol.
$$T_{\frac{2n}{2}+1} = T_{n+1} = {}^{2n}C_n x^n = \frac{(2n)!}{n!(n!)} x^n = \frac{2^n(n!)(1 \cdot 3 \dots (2n-1))}{n! n!} x^n$$

Illustration :

If sum of coefficient in the expansion of $(2 + 3cx + c^2x^2)^{12}$ is zero. Find c .

Sol. Put $x = 1$ in the expansion, then

$$2 + 3c + c^2 = 0$$

$c = -2, c = 1$. Ans.

Illustration :

Find coefficient of x^{15} in $(x - x^2)^{10}$.

Sol. $T_{r+1} = {}^{10}C_r (x)^{10-r} (-1)^r (x^2)^r = {}^{10}C_r (-1)^r (x)^{10+r}$
 $10 + r = 15 : r = 5$ i.e., 6th term
 $T_6 = {}^{10}C_5 (-1)^5 \cdot x^{15} = -{}^{10}C_5 x^{15}$. **Ans.**

Illustration :

Find coefficient term independent of x in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12}$.

Sol. $T_{r+1} = {}^{12}C_r (x^2)^{12-r} \left(\frac{1}{x}\right)^r = {}^{12}C_r (x)^{24-3r}$
 $r = 8$ and 9th term $\Rightarrow T_9$. **Ans.**

Illustration :

Illustration :

Find last four terms in the expansion of $(x + 2x^2)^8$.

Sol. Above is equivalent to first four terms of $(2x^2 + x)^8$
 $T_9 = {}^8C_0 (2x^2)^8 = 256 x^{16}$
 $T_8 = {}^8C_1 (2x^2)^7 x = 1024 x^{15}$
 $T_7 = {}^8C_2 (2x^2)^6 x^2 = 1792 x^{14}$
 $T_6 = {}^8C_3 (2x^2)^5 x^3 = 1792 x^{13}$. **Ans.**

Illustration :

In $\left(2x^2 - \frac{1}{3x}\right)^{11}$ find term involving x^6 also find term independent of x .

Sol. $T_{r+1} = {}^{11}C_r (2x^2)^{11-r} \left(-\frac{1}{3x}\right)^r \cdot \left(\frac{1}{x}\right)^r$
 $= {}^{11}C_r \cdot (2)^{11-r} (-3)^{-r} (x)^{22-3r}$
 $22 - 3r = 6 \Rightarrow r = \frac{16}{3}$ and

For x^6 $22 - 3r = 6 \Rightarrow r = \frac{16}{3}$ (which is not possible)

For independent of x $22 - 3r = 0 \Rightarrow r = \frac{22}{3}$ (which is not possible)

Both do not exist. **Ans.**

Illustration :

In the expansion of $\left(4^{\frac{1}{3}} + \frac{1}{6^{\frac{1}{4}}}\right)^{20}$ find number of rational terms

Sol. $T_{r+1} = {}^{20}C_r \cdot (4)^{\frac{20-r}{3}} \cdot (6)^{\frac{-r}{4}} = {}^{20}C_r \cdot 2^{\frac{40-2r}{3}} \cdot \frac{1}{6^{\frac{r}{4}}} = {}^{20}C_r (2)^{\left\{\frac{160-11r}{12}\right\}} (3)^{\left\{\frac{-r}{4}\right\}}$ $[0 \leq r \leq 20]$

$\therefore r = 8, 20.$

$\therefore$ There are only two rational terms (T_9 and T_{21}) **Ans.**

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$\therefore$ There are only two rational terms (T_9 and T_{21}) **Ans.**

Illustration :

Find greatest value of term independent of x in $\left(x \sin \alpha + \frac{\cos \alpha}{x}\right)^{10}$ $\alpha \in R$.

Sol. $T_{r+1} = {}^{10}C_r (x)^{10-r} \cdot (\sin \alpha)^{10-r} (\cos \alpha)^r \left(\frac{1}{x}\right)^r$

$\Rightarrow T_{r+1} = {}^{10}C_r (x)^{10-2r} (\sin \alpha)^{10-r} (\cos \alpha)^r$

$\Rightarrow 10 - 2r = 0 ; r = 5.$

$\Rightarrow {}^{10}C_5 (\sin \alpha)^5 \cdot (\cos \alpha)^5$

$\Rightarrow T_{r+1} = {}^{10}C_5 \frac{(\sin 2\alpha)^5}{32}$

$\therefore$ Greatest value of $T_{r+1} = \frac{{}^{10}C_5}{32}$. $[\sin 2\alpha = 1]$ **Ans.**

Illustration :

The term independent of x in the expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$ is

- (A) $T_5 = 210$ (B) $T_5 = -210$ (C) $T_4 = 180$ (D) $T_4 = -180$

Sol. We have

$$\begin{aligned}\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10} &= \left(\frac{(x^{1/3})^3+1^3}{x^{2/3}-x^{1/3}+1}-\frac{(x^{1/2})^2-I^2}{x^{1/2}(x^{1/2}-I)}\right)^{10} \\ &= \left((x^{1/3}+I)-\frac{x^{1/2}-I}{x^{1/2}}\right)^{10} = (x^{1/3}+x^{-1/2})^{10}\end{aligned}$$

We have,

$$\begin{aligned}T_{r+1} &= {}^{10}C_r (x^{1/3})^{10-r} (-x^{-1/2})^r \quad \dots\dots\dots(1) \\ &= {}^{10}C_r x^{\frac{10-r}{3}-\frac{r}{2}} (-1)^r\end{aligned}$$

It will be independent of x , if

$$= {}^{10}C_r x^{\frac{10-r}{3}-\frac{r}{2}} (-1)^r$$

It will be independent of x , if

$$\frac{10-r}{3}-\frac{r}{2} = 0.$$

$$\Rightarrow 20-2r-3r=0 \quad \Rightarrow \quad r=4$$

Putting $r=4$ in (1), we get

$$T_5 = {}^{10}C_4 (-1)^4 = {}^{10}C_4 = 210.$$

Hence (a) is correct answer. **Ans.**

Illustration :

The ninth term in the expansion of $\left\{3^{\log_3 \sqrt{25^{x-1}+7}} + 3^{-1/8 \log_3 (5^{x-1}+1)}\right\}^{10}$ is equal to 180, then x is

- (A) 2 (B) 1 (C) 4 (D) None of these

Sol. We have,

$$\left\{3^{\log_3 \sqrt{25^{x-1}+7}} + 3^{-1/8 \log_3 (5^{x-1}+1)}\right\}^{10} = \left[\sqrt{25^{x-1}+7} + (5^{x-1}+1)^{-1/8}\right]^{10} \quad \left\{\because a^{\log_a N} = N\right\}$$

$$\text{Here, } T_9 = 180 \Rightarrow {}^{10}C_8 \left\{\sqrt{25^{x-1}+7}\right\}^{10-8} \left\{(5^{x-1}+1)^{-1/8}\right\} = 180$$

$$\Rightarrow {}^{10}C_8 (25^{x-1} + 7) (5^{x-1} + 1)^{-1} = 180 \quad \Rightarrow \quad 45 \frac{(25^{x-1} + 7)}{5^{x-1} + 1} = 180$$

$$\Rightarrow \frac{25^{x-1} + 7}{5^{x-1} + 1} = 4 \quad \Rightarrow \quad \frac{y^2 + 7}{y + 1} = 4, \text{ where } y = 5^{x-1}$$

$$\Rightarrow y^2 - 4y + 3 = 0 \Rightarrow y = 3, 1 \quad \Rightarrow 5^{x-1} = 3 \text{ or } 5^{x-1} = 1 \Rightarrow 5^x = 15 \text{ or } 5^x = 5.$$

Hence (b) is correct answer. **Ans.**

Illustration :

The coefficient of x^{50} in the expansion

$$(1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \dots + 1001x^{1000} \text{ is}$$

(A) ${}^{1002}C_{50}$

(B) ${}^{1002}C_{51}$

(C) ${}^{1005}C_{50}$

(D) ${}^{1005}C_{48}$

Sol. Let $S = (1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \dots + 1001x^{1000} \quad \dots (1)$

$$xS$$

$$x^{1001}$$

Sol. Let $S = (1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \dots + 1001x^{1000} \quad \dots (1)$

$$\Rightarrow \frac{xS}{1+x} = x(1+x)^{999} + 2x^2(1+x)^{998} + \dots + 1000x^{1000} + 1001 \frac{x^{1001}}{1+x} \quad \dots (2)$$

After subtraction we get

$$\Rightarrow \frac{S}{1+x} = (1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \dots + x^{1000} - 1001 \frac{x^{1001}}{1+x}.$$

$$\Rightarrow \frac{S}{1+x} = (1+x)^{1000} \left(\frac{1 - \left(\frac{x}{1+x} \right)^{1001}}{1 - \frac{x}{1+x}} \right) - 1001 \frac{x^{1001}}{1+x}$$

$$\Rightarrow S = (1+x)^{1002} - x^{1001}(1+x) - 1001x^{1001}$$

$$S = (1+x)^{1002} - 1002x^{1001} - x^{1002}$$

Coefficient of x^{50} in $S = \text{coefficient of } x^{50} \text{ in } (1+x)^{1002} = {}^{1002}C_{50}$. **Ans.**

Illustration :

Find coefficient of x^3 in the expansion of $(1 - x + x^2)^5$.

Sol. $(1 - x + x^2)^5 = \{1 + x(x-1)\}^5 = {}^5C_0 + {}^5C_1x(x-1) + {}^5C_2x^2(x-1)^2 + {}^5C_3x^3(x-1)^3 + \dots$
Coefficient of $x^3 = -2^5C_2 - {}^5C_3 = -20$. **Ans.**

Illustration :

Find coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^{11}$.

Sol. $(1 + x + x^2 + x^3)^{11} = (1 + x)^{11} (1 + x^2)^{11}$
 $= (1 + {}^{11}C_1x + {}^{11}C_2x^2 + {}^{11}C_3x^3 + {}^{11}C_4x^4 + \dots) \times (1 + {}^{11}C_1x^2 + {}^{11}C_2x^4 + \dots)$
(The terms which gives x^4 are)
 $= {}^{11}C_2 + ({}^{11}C_2 \times {}^{11}C_1) + {}^{11}C_4 = 55 + 605 + 330 = 990$. **Ans.**

Illustration :

If $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$ ($n \in N$), then

- | | |
|--|--|
| (a) $a_0 + a_1 + a_2 + a_3 + \dots + a_{2n}$ | (b) $a_0 - a_1 + a_2 - a_3 + \dots + a_{2n}$ |
| (c) $a_1 + a_3 + a_5 + \dots + a_{2n-1}$ | (d) $a_0 + a_2 + a_4 + \dots + a_{2n}$ |
| (e) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2$ | (f) $a_0a_1 - a_1a_2 + a_2a_3 + \dots$ |
| (g) $a_0a_2 - a_1a_3 + a_2a_4 + \dots$ | |

Sol.

(a) $a_0 + a_1 + a_2 + a_3 + \dots + a_{2n} = 3^n$ put $x = 1$

(b) $a_0 - a_1 + a_2 - a_3 + \dots + a_{2n} = 1$ put $x = -1$

(c) From (a) and (b) $a_1 + a_3 + a_5 + \dots + a_{2n-1} = \frac{3^n - 1}{2}$

(d) From (a) and (b) $a_0 + a_2 + a_4 + \dots + a_{2n} = \frac{3^n + 1}{2}$

(e) $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$... (i)

$$\left(1 - \frac{1}{x} + \frac{1}{x^2}\right)^n = a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} + \dots + \frac{a_{2n}}{x^{2n}}$$

$$\Rightarrow \frac{1}{x^{2n}} (x^2 - x + 1)^n = a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} + \dots + \frac{a_{2n}}{x^{2n}} \quad \dots (ii)$$

From (i) & (ii)

$$(1 + x + x^2)^n \left(1 - \frac{1}{x} + \frac{1}{x^2}\right)^n = (a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}) \left(a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} + \dots + \frac{a_{2n}}{x^{2n}}\right)$$

$$\Rightarrow \frac{1}{x^{2n}} (x^4 + x^2 + 1)^n = (a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}) \left(a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} + \dots + \frac{a_{2n}}{x^{2n}}\right) \dots (iii)$$

Compairing constant terms on both sides.

$$a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2 = a_n$$

(f) Compairing coefficient of x in equation (iii) on both sides.

$$a_0a_1 - a_1a_2 + a_2a_3 + \dots = 0$$

(g) Compairing coefficient of x^2 in equation (iii) on both sides.

$$a_0a_2 - a_1a_3 + a_2a_4 + \dots = a_{n+1} \text{ or } a_{n-1}$$

Practice Problem

Practice Problem

Q.1 Find coefficient of x^{20} in the expansion of $(1 + x^2)^{40} \left(x^2 + 2 + \frac{1}{x^2}\right)^{-5}$.

Q.2 Find term independent of x in $\left[\sqrt{\frac{x}{3}} + \sqrt{\frac{3}{2x^2}}\right]^{10}$.

Q.3 Find the ratio of coefficients of x^{10} in $(1 - x^2)^{10}$ and term independent of x in $\left(x - \frac{2}{x}\right)^{10}$

Answer key

Q.1 $^{30}C_{25}$

Q.2 No term

Q.3 $\frac{1}{32}$

3.1 NUMERICALLY GREATEST TERM IN $(x + y)^n$:

T_{r+1} term is said to be numerically greatest for a given value of x, y provided $T_{r+1} \geq T_r$ and

$$T_{r+1} \geq T_{r+2} \Rightarrow \frac{T_{r+1}}{T_r} \geq 1 \text{ as well as } \frac{T_{r+1}}{T_{r+2}} \geq 1$$

$$\frac{T_{r+1}}{T_r} = \frac{{}^nC_r (x)^{n-r} y^r}{{}^nC_{r-1} (x)^{n-r+1} y^{r-1}} = \frac{(n-r+1)}{r} \left[\frac{y}{x} \right]$$

Numerically
greatest term
is required

Illustration :

Find the greatest term in the expansion of $(7 - 5x)^{11}$ where $x = 2/3$.

Sol. Since we have to find numerically greatest term,

$$\therefore \text{greatest term in } (7 - 5x)^{11} = \text{greatest term in } (7 + 5x)^{11}$$

Let r th term be the greatest term in the expansion of $(7 + 5x)^{11}$

$$\text{Now, } t_r = {}^{11}C_{r-1} (7)^{11-r+1} (5x)^{r-1} = {}^{11}C_{r-1} 7^{12-r} (5x)^{r-1} \quad \dots(i)$$

$$\text{and } (r+1)\text{th term, } t_{r+1} = {}^{11}C_r (7)^{11-r} (5x)^r \quad \dots(ii)$$

$$\therefore \frac{t_r}{t_{r+1}} = \frac{{}^{11}C_{r-1} \cdot 7^{12-r} (5x)^{r-1}}{{}^{11}C_r \cdot 7^{11-r} (5x)^r}$$

$$\text{or } \frac{t_r}{t_{r+1}} = \frac{21r}{(12-r)10} \geq 1 \quad \dots(iii)$$

$$\text{or } 21r \geq 120 - 10r \quad \text{or } r \geq 3 \frac{27}{31}$$

$$\text{Putting } (r-1) \text{ in place of } r \text{ in (iii), } \frac{t_{r-1}}{t_r} = \frac{21(r-1)}{\{12-(r-1)\} \cdot 10} = \frac{21r-21}{130-10r} \leq 1$$

$$\text{or } r \leq 4 \frac{27}{31} \quad \dots (iv)$$

From (iii) and (iv) $r = 4$

$$\therefore \text{Greatest term} = t_4 = {}^{11}C_3 \cdot 7^8 \left(5 \cdot \frac{2}{3}\right)^3 = \frac{440}{9} \cdot 7^8 \cdot 5^3$$

Illustration :

Given T_4 in the expansion of $\left(2 + \frac{3x}{8}\right)^{10}$ has maximum numerical value find range of x .

Sol. $T_4 > T_3$ and $T_4 > T_5$

$$\Rightarrow \frac{10-3+1}{3} \left| \frac{3x}{8 \times 2} \right| \quad \text{and} \quad \frac{10-4+1}{4} \left| \frac{3x}{16} \right| < 1$$

$$\Rightarrow |x| > 2 \quad \text{and} \quad |x| < \frac{64}{21}$$

$$\Rightarrow x > 2 ; x < -2 \quad \text{and} \quad \frac{-64}{x} < x < \frac{64}{21}$$

$$\overbrace{x \in \left(\frac{-64}{21}, -2\right) \cup \left(2, \frac{64}{21}\right)}^{\text{Inter section}}$$

$$\overbrace{x \in \left(\frac{-64}{21}, -2\right) \cup \left(2, \frac{64}{21}\right)}$$

Illustration :

Find the index n of the binomial $\left(\frac{x}{5} + \frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically the greatest coefficient ($n \in N$) (at $x = 1$)

Sol. If T_9 is greatest term then

$$\frac{T_9}{T_8} > 1 \quad \text{and} \quad \frac{T_{10}}{T_9} < 1$$

$$\left(\frac{n-8+1}{8}\right) \frac{2}{5} \cdot 5 > 1 \quad \text{and} \quad \frac{2}{5} \cdot 5 \left(\frac{n-9+1}{9}\right) < 1$$

$$2n - 14 > 8 \quad \text{and} \quad 2n - 16 < 9$$

$$n > 11 \quad \text{and} \quad 11 < n < 12.5$$

$$\therefore n = 12. \text{ Ans.}$$

3.2 BINOMIAL COEFFICIENTS & THEIR PROPERTIES :

Properties of nC_r

$$(1) \quad {}^nC_r = {}^nC_{n-r} \Rightarrow {}^nC_x = {}^nC_y \text{ has two solutions } x = y \text{ or } x + y = n.$$

$$(2) \quad {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$$(3) \quad {}^nC_r = \frac{n}{r} ({}^{n-1}C_{r-1})$$

$$(4) \quad \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}.$$

Note :

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\int_a^b x^n dx = \left[\frac{x^{n+1}}{n+1} \right]_a^b = \frac{b^{n+1} - a^{n+1}}{n+1}$$

In the expansion of $(1+x)^n$; i.e. $(1+x)^n = {}^nC_0 + {}^nC_1x + \dots + {}^nC_r x^r + \dots + {}^nC_n x^n$

The coefficients ${}^nC_0, {}^nC_1, {}^nC_n$ of various powers of x , are called binomial coefficients and they are written as

$$C_0, C_1, C_2, \dots, C_n$$

Hence

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_r x^r + \dots + C_n x^n \quad \dots(1)$$

$$\text{Where } C_0 = 1, C_1 = n, C_2 = \frac{n(n-1)}{2!}$$

$$C_r = \frac{n(n-1)\dots(n-r+1)}{r!}, \quad C_n = 1$$

Now, we shall obtain some important expressions involving binomial coefficients-

(a) **Sum of Coefficient :** putting $x = 1$ in (1), we get

$$C_0 + C_1 + C_2 + \dots + C_n = 2^n \quad \dots(2)$$

- (b) **Sum of coefficients with alternate signs :** putting $x = -1$ in (1)

We get

$$C_0 - C_1 + C_2 - C_3 + \dots = 0 \quad \dots(3)$$

- (c) **Sum of coefficients of even and odd terms:** from (3), we have

$$C_0 + C_2 + C_4 + \dots = C_1 + C_3 + C_5 + \dots \quad \dots(4)$$

i.e. sum of coefficients of even and odd terms are equal.

from (2) and (4)

$$\Rightarrow C_0 + C_2 + \dots = C_1 + C_3 + \dots = 2^{n-1}$$

- (d) **Sum of products of coefficients :** Replacing x by $1/x$ in (1)

We get

$$\left(1 + \frac{1}{x}\right)^n = C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \dots + \frac{C_n}{x^n} + \dots \quad \dots(5)$$

Multiplying (1) by (5), we get

$$\frac{(1+x)^{2n}}{x^n} = (C_0 + C_1x + C_2x^2 + \dots) \left(C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \dots\right)$$

Now, comparing coefficients of x^r on both the sides, we get

$$\begin{aligned} C_0C_r + C_1C_{r+1} + \dots + C_{n-r}C_n &= {}^{2n}C_{n-r} \\ &= \frac{2n!}{(n+r)!(n-r)!} \quad \dots(6) \end{aligned}$$

- (e) **Sum of squares of coefficients :**

putting $r = 0$ in (6), we get

$$C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = \frac{2n!}{n!n!}$$

- (f) putting $r = 1$ in (6), we get

$$\begin{aligned} C_0C_1 + C_1C_2 + C_2C_3 + \dots + C_{n-1}C_n &= {}^{2n}C_{n-1} \\ &= \frac{2n!}{(n+1)!(n-1)!} \quad \dots(7) \end{aligned}$$

- (g) putting $r = 2$ in (6), we get

$$\begin{aligned} C_0C_2 + C_1C_3 + C_2C_4 + \dots + C_{n-2}C_n &= {}^{2n}C_{n-2} \\ &= \frac{2n!}{(n+2)!(n-2)!} \quad \dots(8) \end{aligned}$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, prove that $C_0^2 - C_1^2 + C_2^2 - \dots + (-1)^n \cdot (-1)^{n/2} C_n^2 = 0$

or $(-1)^{n/2} \frac{n!}{(n/2)!(n/2)!}$ according as n is odd or even.

Sol. Since

$$(1-x)^n = C_0 - C_1x + C_2x^2 - \dots + (-1)^n C_nx^n \quad \dots(i)$$

$$\text{and } (x+1)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \dots + C_n \quad \dots(ii)$$

Multiplying (i) and (ii), we get

$$(1-x^2)^n = (C_0 - C_1x + C_2x^2 - \dots + (-1)^n C_nx^n) (C_0x^n + C_1x^{n-1} + C_2x^{n-2} + \dots + C_n) \quad \dots(iii)$$

Now, coefficient of x^n in R.H.S.

$$= C_0^2 - C_1^2 + C_2^2 - \dots + (-1)^n C_n^2$$

$$\text{General term in L.H.S.} = T_{r+1} = {}^nC_r (-x^2)^r$$

$$= {}^nC_r (-1)^r x^{2r}$$

$$\text{Putting } 2r = n$$

$$\therefore r = n/2$$

$$\therefore T_{(n/2)+1} = {}^nC_{n/2} (-1)^{n/2} x^n$$

$$\therefore \text{Coefficient of } x^n \text{ in L.H.S.} = {}^nC_{n/2} (-1)^{n/2}$$

$$= (-1)^{n/2} \frac{n!}{(n/2)!(n/2)!} = \begin{cases} 0 & \text{if } n \text{ is odd} \\ (-1)^{n/2} \frac{n!}{(n/2)!(n/2)!} & \text{if } n \text{ is even} \end{cases}$$

But (iii) is an identity, therefore coefficient of x^n in R.H.S. = coefficient of x^n in L.H.S.

$$\Rightarrow C_0^2 - C_1^2 + C_2^2 - \dots + (-1)^n C_n^2 = \begin{cases} 0 & \text{if } n \text{ is odd} \\ (-1)^{n/2} \frac{n!}{(n/2)!(n/2)!} & \text{if } n \text{ is even} \end{cases}$$

Illustration :

Prove that ${}^{m+n}C_r = {}^mC_r + {}^mC_{r-1} {}^nC_1 + {}^mC_{r-2} {}^nC_2 + \dots + {}^nC_r$ if $r < m$, $r < n$ and m, n, r are positive integers.

Sol. Here sum of lower suffices of binomial coefficient in each term is r

$$\text{i.e. } r = r-1+1 = r-2+2 = \dots = r = r$$

$$\text{since } (1+x)^m = {}^mC_0 + {}^mC_1x + \dots + {}^mC_{r-2}x^{r-2} + {}^mC_{r-1}x^{r-1} + {}^mC_rx^r + \dots + {}^mC_mx^m \quad \dots(i)$$

$$\text{and } (1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_rx^r + \dots + {}^nC_nx^n \quad \dots(ii)$$

multiplying (i) and (ii), we get

$$(1+x)^{m+n} = ({}^mC_0 + {}^mC_1x + \dots + {}^mC_{r-2}x^{r-2} + {}^mC_{r-1}x^{r-1} + {}^mC_rx^r + \dots + \dots + {}^mC_mx^m) \times ({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_rx^r + \dots + {}^nC_nx^n) \quad \dots(iii)$$

Now coefficient of x^r in R.H.S.

$$\begin{aligned} &= {}^mC_r \cdot {}^nC_0 + {}^mC_{r-1} \cdot {}^nC_1 + {}^mC_{r-2} \cdot {}^nC_2 + \dots + {}^mC_0 \cdot {}^nC_r \\ &= {}^mC_r + {}^mC_{r-1} \cdot {}^nC_1 + {}^mC_{r-2} \cdot {}^nC_2 + \dots + {}^nC_r \end{aligned}$$

Coefficient of x^r in L.H.S. $= {}^{m+n}C_r$

But (iii) is an identity, therefore coefficient of x^r in L.H.S. = coefficient of x^r R.H.S.

$${}^{m+n}C_r = {}^mC_r + {}^mC_{r-1} \cdot {}^nC_1 + {}^mC_{r-2} \cdot {}^nC_2 + \dots + {}^nC_r$$

(h) Use of Differentiation :

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_rx^r + \dots + C_nx^n \quad \dots(1)$$

Differentiating both sides of (1) w.r.t. x , we get

$$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \dots + nC_nx^{n-1}$$

Now putting $x = 1$ and $x = -1$ respectively

$$C_1 + 2C_2 + 3C_3 + \dots + nC_n = n \cdot 2^{n-1} \quad \dots(9)$$

Now putting $x = 1$ and $x = -1$ respectively

$$C_1 + 2C_2 + 3C_3 + \dots + nC_n = n \cdot 2^{n-1} \quad \dots(9)$$

$$\text{and } C_1 - 2C_2 + 3C_3 - \dots = 0 \quad \dots(10)$$

(i) Adding (2) and (9)

$$C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = 2^{n-1}(n+2) \quad \dots(11)$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, then prove that
 $C_1 + 2C_2 + 3C_3 + \dots + nC_n = n \cdot 2^{n-1}$.

Sol. Given series is

$$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$$

then differentiating both sides w.r. to x , we get

$$\Rightarrow n(1+x)^{n-1} = 0 + C_1 + 2C_2x + 3C_3x^2 + \dots + nC_nx^{n-1}$$

Putting $x = 1$, we get

$$n \cdot 2^{n-1} = C_1 + 2C_2 + 3C_3 + \dots + nC_n$$

$$\text{or } C_1 + 2C_2 + 3C_3 + \dots + nC_n = n \cdot 2^{n-1}$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, then prove that

$$C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = (n+2)2^{n-1}.$$

Sol. Given series is

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$$

Now replacing by x^1 and multiplying both sides by x^1 , then

$$x(1+x)^n = C_0x + C_1x^2 + C_2x^3 + \dots + C_nx^{n+1}$$

Now differentiating both sides w.r.t. x , we get

$$x \cdot n(1+x)^{n-1} + (1+x)^n \cdot 1 = C_0 + 2C_1x + 3C_2x^2 + \dots + (n+1)C_nx^n$$

Putting $x = 1$, we get

$$n(2)^{n-1} + 2^n = C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n$$

$$\text{or } C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = (n+2)2^{n-1}$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ then prove that

$$C_0 + 3C_1 + 5C_2 + \dots + (2n+1)C_n = (n+1)2^n$$

Sol. The given series is

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$$

Now replacing x by x^2

$$\text{then } (1+x^2)^n = C_0 + C_1x^2 + C_2x^4 + \dots + C_nx^{2n}$$

multiplying both sides by x^1 , we get

$$x(1+x^2)^n = C_0x + C_1x^3 + C_2x^5 + \dots + C_nx^{2n+1}$$

then differentiating both sides w.r. to x , we get

$$x \cdot n(1+x^2)^{n-1} \cdot 2x + (1+x^2)^n \cdot 1 = C_0 + 3C_1x^2 + 5C_2x^4 + \dots + (2n+1)C_nx^{2n}$$

Putting $x = 1$, then we get

$$n \cdot 1^{n-1} \cdot 2 + 2^n = C_0 + 3C_1 + 5C_2 + \dots + (2n+1)C_n$$

$$\text{or } C_0 + 3C_1 + 5C_2 + \dots + (2n+1)C_n = (n+1)2^n$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ then prove that

$$(1.2)C_2 + (2.3)C_3 + \dots + ((n-1) \cdot n)C_n = n(n-1)2^{n-2}$$

Sol. The given series is

$$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$$

Differentiating both sides w.r. to x , we get

$$n(1+x)^{n-1} = 0 + C_1 + 2C_2x + 3C_3x^2 + \dots + nC_nx^{n-1}$$

again differentiating both sides w.r. to x , we get

$$n(n-1)(1+x)^{n-2} = 0 + 0 + (1.2)C_2 + (2.3)C_3x + \dots + ((n-1).n)C_nx^{n-2}$$

Putting $x = 1$, then

$$n(n-1)(1+1)^{n-2} = (1.2)C_2 + (2.3)C_3 + \dots + (n-1)n C_n$$

$$\text{or } (1.2)C_2 + (2.3)C_3 + \dots + (n-1)n \cdot C_n = n(n-1) 2^{n-2}.$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$ then prove that

$$C_0 + 2C_1 + 3C_2 - 4C_3 + \dots + (-1)^n (n+1) C_n = 0$$

Sol. The given series is

$$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$$

Multiplying both sides by x , then

$$x(1+x)^n = C_0x + C_1x^2 + C_2x^3 + C_3x^4 + \dots + C_nx^{n+1}$$

$$x(1+x)^n = C_0x + C_1x^2 + C_2x^3 + C_3x^4 + \dots + C_nx^{n+1}$$

Differentiating both sides w.r. to x , then we get

$$x.n(1+x)^{n-1} + (1+x)^n \cdot 1 = C_0 + 2C_1x + 3C_2x^2 + 4C_3x^3 + \dots + (n-1)C_nx^n$$

Putting $x = -1$, then we get

$$0 = C_0 - 2C_1 + 3C_2 - 4C_3 + \dots + (-1)^n (n+1) C_n$$

$$\text{or } C_0 - 2C_1 + 3C_2 - 4C_3 + \dots + (-1)^n (n+1)C_n = 0$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$ then prove that

$$C_1 - 2C_2 + 3C_3 - \dots + (-1)^{n-1} n C_n = 0$$

Sol. $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$

Differentiating both sides w.r. to x , we get

$$n(1+x)^{n-1} = 0 + C_1 + 2C_2x + 3C_3x^2 + \dots + nC_nx^{n-1}$$

Putting $x = -1$, then we get

$$0 = C_1 - 2C_2 + 3C_3 - \dots + (-1)^{n-1} n C_n$$

$$\text{or } C_1 - 2C_2 + 3C_3 - \dots + (-1)^{n-1} n C_n = 0$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$, then prove that

$$C_0 - 3C_1 + 5C_2 - \dots + (-1)^n (2n+1) C_n = 0$$

Sol. The given series is

$$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n \text{ replacing } x \text{ by } x^2 \text{ then}$$

$$(1+x^2)^n = C_0 + C_1x^2 + C_2x^4 + \dots + C_nx^{2n}$$

Multiplying both sides by x , then

$$x(1+x^2)^n = C_0x + C_1x^3 + C_2x^5 + \dots + C_nx^{2n+1}$$

Differentiating both sides w.r. to x , we get

$$x.n(1+x^2)^{n-1} \cdot 2n + (1+x^2)^n \cdot 1 = C_0 + 3C_1x^2 + 5C_2x^4 + \dots + (2n+1)C_nx^{2n}$$

Putting $x = i$ in both sides, we get

$$0 + 0 = C_0 - 3C_1 + 5C_2 - \dots + (2n+1)(-1)^n C_n$$

or
$$C_0 - 3C_1 + 5C_2 - \dots + (-1)^n (2n+1) C_n = 0$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$ then prove that

$$C_1^2 + 2C_2^2 + 3C_3^2 + \dots + nC_n^2 = \frac{(2n-1)!}{((n-1)!)^2}$$

$$C_1^* + 2C_2^* + 3C_3^* + \dots + nC_n^* = \frac{(2n-1)!}{((n-1)!)^2}$$

Sol. Given $(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$

Differentiating both sides w.r.t. to x , we get

$$n(1+x)^{n-1} = 0 + C_1 + 2C_2x + 3C_3x^2 + \dots + nC_nx^{n-1}$$

$$\Rightarrow n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \dots + nC_nx^{n-1} \quad \dots(i)$$

$$\text{and } (x+1)^n = C_0x^n + C_1x^{n-1} + C_2x^{n-2} + C_3x^{n-3} + \dots + C_n \quad \dots(ii)$$

Multiplying (i) and (ii), we get

$$n(1+x)^{2n-1} = (C_1 + 2C_2x + 3C_3x^2 + \dots + nC_nx^{n-1}) \times (C_0x^n + C_1x^{n-1} + C_2x^{n-2} + C_3x^{n-3} + \dots + C_n) \quad \dots(iii)$$

Now, coefficient of x^{n-1} on R.H.S.

$$C_1^2 + 2C_2^2 + 3C_3^2 + \dots + nC_n^2$$

and coefficient of x^{n-1} on L.H.S. = $n \cdot {}^{2n-1}C_{n-1}$

$$= n \cdot \frac{(2n-1)!}{(n-1)!n!} = \frac{(2n-1)!}{(n-1)!(n-1)!} = \frac{(2n-1)!}{\{(n-1)!\}^2}$$

but (iii) is an identity, therefore the coefficient of x^{n-1} in R.H.S. = coefficient of x^{n-1} in L.H.S.

$$\Rightarrow C_1^2 + 2C_2^2 + 3C_3^2 + \dots + nC_n^2 = \frac{(2n-1)!}{\{(n-1)!\}^2}$$

(j) **Use of Integration :**

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_r x^r + \dots + C_n x^n \quad \dots(1)$$

Integrating (1) w.r.t. x between the limits 0 to 1, we get,

$$\left[\frac{(1+x)^{n+1}}{n+1} \right]_0^1 = \left[C_0x + C_1 \frac{x^2}{2} + C_2 \frac{x^3}{3} + \dots + \frac{C_n x^{n+1}}{n+1} \right]$$

$$\Rightarrow C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1} \quad \dots(12)$$

Integrating (1) w.r.t. x between the limits -1 to 0, we get

$$\left[\frac{(1+x)^{n+1}}{n+1} \right]_{-1}^0 = \left[C_0x + C_1 \frac{x^2}{2} + C_2 \frac{x^3}{3} + \dots + \frac{C_n x^{n+1}}{n+1} \right]_{-1}^0$$

$$\Rightarrow C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} + \dots + \frac{(-1)^n \cdot C_n}{n+1} = \frac{1}{(n+1)} \quad \dots(13)$$

Illustration :

If $C_r = {}^nC_r$, then prove that $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$.

Sol. Consider the expansion

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_n x^n \quad \dots(i)$$

Integrating both sides of (i) within limits 0 to 1, we get

$$\int_0^1 (1+x)^n dx = \int_0^1 (C_0 + C_1x + C_2x^2 + \dots + C_n x^n) dx$$

$$\Rightarrow \left[\frac{(1+x)^{n+1}}{n+1} \right]_0^1 = \left[C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + \dots + \frac{C_n x^{n+1}}{n+1} \right]_0^1$$

$$\Rightarrow \frac{2^{n+1} - 1}{n+1} = C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1}$$

$$\text{Hence } C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$$

Illustration :

Prove that $C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \dots + (-1)^n \frac{C_n}{n+1} = \frac{1}{n+1}$

Sol. Consider the expansion $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$
Integrating both sides of (i) within limits -1 to 0 , we get

$$\begin{aligned}\int_{-1}^0 (1+x)^n dx &= \int_{-1}^0 (C_0 + C_1x + C_2x^2 + \dots + C_nx^n) dx \\ \Rightarrow \left[\frac{(1+x)^{n+1}}{n+1} \right]_{-1}^0 &= \left[C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + \dots + \frac{C_nx^{n+1}}{n+1} \right]_{-1}^0 \\ \Rightarrow \frac{1-0}{n+1} &= 0 - \left(-C_0 + \frac{C_1}{2} - \frac{C_2}{3} + \dots + (-1)^{n+1} \frac{C_n}{n+1} \right) \\ \Rightarrow \frac{1}{n+1} &= C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \dots + (-1)^n \frac{C_n}{n+1} \\ \text{Hence } C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \dots + (-1)^n \frac{C_n}{n+1} &= \frac{1}{n+1}\end{aligned}$$

Illustration :

Prove that $\frac{C_0}{1} + \frac{C_2}{3} + \frac{C_4}{5} + \dots = \frac{2^n}{n+1}$

Sol. Consider the expansion

$$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + \dots + C_nx^n \dots (i)$$

Integrating both sides of (i) within limits -1 to 1 , we get

$$\begin{aligned}\int_{-1}^1 (1+x)^n dx &= \int_{-1}^1 (C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + \dots + C_nx^n) dx \\ &= \int_{-1}^1 (C_0 + C_2x^2 + C_4x^4 + \dots) dx + \int_{-1}^1 (C_1x + C_3x^3 + \dots) dx \\ &= 2 \int_{-1}^1 (C_0 + C_2x^2 + C_4x^4 + \dots) dx + 0 \quad (\text{By prop. of definite integral}) \\ &\quad (\text{since second integral contains odd function})\end{aligned}$$

$$\Rightarrow \left[\frac{(1+x)^{n+1}}{n+1} \right]_{-1}^1 = 2 \left[C_0x + \frac{C_2x^3}{3} + \frac{C_4x^5}{5} + \dots \right]_{-1}^1$$

$$\Rightarrow \frac{2^{n+1}}{n+1} = 2 \left(C_0 + \frac{C_2}{3} + \frac{C_4}{5} + \dots \right)$$

$$\text{Hence } C_0 + \frac{C_2}{3} + \frac{C_4}{5} + \dots = \frac{2^n}{n+1}$$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, prove that

$$3C_0 + 3^2 \frac{C_1}{2} + \frac{3^3 C_2}{3} + \frac{3^4 C_3}{4} + \dots + \frac{3^{n+1} C_n}{n+1} = \frac{4^{n+1} - 1}{n+1}$$

Sol. Consider the expansion

$$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$$

Integrating both sides of (i) within limits 0 to 3, we get :

$$\int_0^3 (1+x)^n dx = \int_0^3 (C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n) dx$$

$$\Rightarrow \left[\frac{(1+x)^{n+1}}{n+1} \right]_0^3 = \left[C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + \frac{C_3x^4}{4} + \dots + \frac{C_nx^{n+1}}{n+1} \right]_0^3$$

$$\Rightarrow \frac{4^{n+1} - 1}{n+1} = 3C_0 + \frac{3^2 C_1}{2} + \frac{3^3 C_2}{3} + \frac{3^4 C_3}{4} + \dots + \frac{3^{n+1} C_n}{n+1}$$

$$\text{Hence } 3C_0 + \frac{3^2 C_1}{2} + \frac{3^3 C_2}{3} + \frac{3^4 C_3}{4} + \dots + \frac{3^{n+1} C_n}{n+1} = \frac{4^{n+1} - 1}{n+1}$$

Illustration :

Prove that ${}^{15}C_4 + 2{}^{15}C_5 + {}^{15}C_6 = {}^{17}C_6$

Sol. ${}^{15}C_4 + 2{}^{15}C_5 + {}^{15}C_6 = ({}^{15}C_4 + {}^{15}C_5) + ({}^{15}C_5 + {}^{15}C_6) = {}^{16}C_5 + {}^{16}C_6 = {}^{17}C_6$ **Ans.**

Illustration :

$$\left(\frac{{}^nC_0 + {}^nC_1}{{}^nC_0} \right) \left(\frac{{}^nC_1 + {}^nC_2}{{}^nC_1} \right) \dots \left(\frac{{}^nC_{n-1} + {}^nC_n}{{}^nC_{n-1}} \right) = \frac{(15)^{14}}{14!} \text{ find value of } n.$$

Sol. Given product series may be written as $\prod_{r=1}^n \left(\frac{{}^nC_{r-1} + {}^nC_r}{{}^nC_{r-1}} \right)$

$$= \prod_{r=1}^n \frac{{}^{n+1}C_r}{{}^nC_{r-1}} = \prod_{r=1}^n \frac{n+1}{r} = \frac{(n+1)}{1} \times \frac{(n+1)}{2} \times \frac{(n+1)}{3} \dots \frac{(n+1)}{n} = \frac{(n+1)^n}{n!} \Rightarrow n = 14. \text{ Ans.}$$

Illustration :

Find the sum of $S = {}^nC_0 + 2({}^nC_1) + 3({}^nC_2) + \dots + (n+1)({}^nC_n)$.

Sol. Given sum may be written as

$$\begin{aligned} S &= \sum_{r=1}^n (r+1)({}^nC_r) = \sum_{r=0}^n r \cdot ({}^nC_r) + \sum_{r=0}^n ({}^nC_r) = \sum_{r=0}^n n \cdot ({}^{n-1}C_{r-1}) + 2^n \\ &= n \sum_{r=0}^n {}^{n-1}C_{r-1} + 2^n = n \cdot 2^{n-1} + 2^n. \text{ Ans.} \end{aligned}$$

Alternate Method :

Consider $x(1+x)^n = {}^nC_0 x + {}^nC_1 x^2 + \dots + {}^nC_n x^{n+1}$

Differtentate w.r.t. x $(1+x)^n + nx(1+x)^{n-1} = {}^nC_0 + 2({}^nC_1)x + \dots + (n+1)({}^nC_n)x^n$

Put $x = 1$ to get ${}^nC_0 + 2({}^nC_1) + 3({}^nC_2) + \dots + (n+1)({}^nC_n) = n \cdot 2^{n-1} + 2^n$
Ans.

Illustration :

Find value of sum $\frac{{}^nC_0}{2} + \frac{{}^nC_1}{3} + \frac{{}^nC_2}{4} \dots$ up to n term

Find value of sum $\frac{{}^nC_0}{2} + \frac{{}^nC_1}{3} + \frac{{}^nC_2}{4} \dots$ up to n term

Sol. $S = \frac{{}^nC_0}{2} + \frac{{}^nC_1}{3} + \frac{{}^nC_2}{4} \dots$

$$= \sum_{r=0}^n \frac{{}^nC_r}{r+2}$$

Consider $x(1+x)^n = {}^nC_0 x + {}^nC_1 x^2 + {}^nC_2 x^3 + \dots + {}^nC_n x^{n+1}$

Intergrate both side w.r.t x

$$\int_0^1 x(1+x)^n dx = \int_0^1 ({}^nC_0 x + {}^nC_1 x^2 + \dots + {}^nC_n x^{n+1}) dx$$

$$\text{or} \quad \left[\frac{x \cdot (1+x)^{n+1}}{n+1} \right]_0^1 - \int_0^1 \frac{(1+x)^{n+1}}{n+1} dx = \left[{}^nC_0 \frac{x^2}{2} + {}^nC_1 \frac{x^3}{3} \dots \right]_0^1$$

$$\text{or} \quad \frac{2^{n+1}}{n+1} - \left(\frac{2^{n+2}}{(n+1)(n+2)} - \frac{1}{(n+1)(n+2)} \right) = \frac{{}^nC_0}{2} + \frac{{}^nC_1}{3} + \dots + \frac{{}^nC_n}{n+2}$$

$$\Rightarrow S = \frac{2^{n+1}}{n+1} - \frac{2^{n+2}}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)}$$

Illustration :

Find sum of series $S = {}^nC_1 + 2 \cdot {}^nC_2 + 3 \cdot {}^nC_3 + \dots + n \cdot {}^nC_n$

Sol.
$$S = \sum_{r=1}^n r \cdot {}^nC_r = \sum_{r=1}^n r \cdot \frac{n}{r} \cdot {}^{n-1}C_{r-1} = n \sum_{r=1}^n {}^{n-1}C_{r-1}$$
$$= n \cdot 2^{n-1}$$

Alternate Method :

$$S = 0 \cdot {}^nC_0 + 1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 + \dots + n \cdot {}^nC_n$$
$$S = n \cdot {}^nC_n + (n-1) \cdot {}^nC_{n-1} + \dots + 0 \cdot {}^nC_0$$
$$\Rightarrow 2S = n \cdot {}^nC_0 + n \cdot {}^nC_1 + \dots + n \cdot {}^nC_n$$

or
$$S = \frac{n}{2} ({}^nC_0 + {}^nC_1 + \dots + {}^nC_n)$$

$$\Rightarrow S = \frac{n}{2} 2^n$$

An Important Result :

An Important Result :

For sums involving product of two binomial coefficients use

$${}^nC_0 {}^mC_k + {}^nC_1 {}^mC_{k-1} + {}^nC_2 {}^mC_{k-2} + \dots + {}^nC_k {}^mC_0 = {}^{m+n}C_k$$

Proof: Consider two series

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_k x^k + \dots + {}^nC_n x^n \quad \dots(1)$$

$$(1+x)^m = {}^mC_m x^m + \dots + {}^mC_k x^k + {}^mC_{k-1} x^{k-1} + \dots + {}^mC_0 \quad \dots(2)$$

Here second series is written in reverse order.

Multiplying (1) and (2) and equate coefficients of x^k on both sides to get

$${}^{m+n}C_k = {}^nC_0 {}^mC_k + {}^nC_1 {}^mC_{k-1} + \dots + {}^nC_k {}^mC_0.$$

Illustration :

Find the sum $S = {}^nC_0 {}^nC_1 + {}^nC_1 {}^nC_2 + \dots + {}^nC_{n-1} {}^nC_n$.

Sol.
$$S = {}^nC_0 {}^nC_{n-1} + {}^nC_1 {}^nC_{n-2} + \dots + {}^nC_{n-1} {}^nC_0 \quad [{}^nC_r = {}^nC_{n-r}]$$

If sum of lower suffices is constant then

$$\Rightarrow S = {}^{n+n}C_{n-1} = {}^{2n}C_{n-1} = {}^{2n}C_{n+1} \text{ Ans.}$$

Illustration :

Find the sum $S = 1 \cdot ({}^nC_0)^2 + 2 \cdot ({}^nC_1)^2 + 3 \cdot ({}^nC_2)^2 + \dots + (n+1) ({}^nC_n)^2$

Sol. Given sum may be written as

$$S = \sum_{r=1}^n r \cdot ({}^nC_r)^2 \quad \text{or} \quad S = \sum_{r=1}^n (r \cdot {}^nC_r) {}^nC_r \quad \text{or} \quad S = \sum_{r=1}^n n \cdot {}^{n-1}C_{r-1} \cdot {}^nC_r$$

$$\text{or} \quad S = n \sum_{r=1}^n {}^{n-1}C_{r-1} \cdot {}^nC_r = n \sum_{r=1}^n {}^{n-1}C_{r-1} {}^nC_{n-r} \Rightarrow S = n \cdot {}^{2n-1}C_{n-1} \cdot \text{Ans.}$$

Illustration :

If $(1+x)^n = C_0 + C_1x + \dots + C_nx^n$, then the value of $\sum_{r=0}^n \sum_{s=0}^n (C_r + C_s)$ is equal to

- (A) $(n+1) 2^{n+1}$ (B) $(n-1)2^{n+1}$ (C) $(n+1) 2^n$ (D) none of these

Sol. We have, $\sum_{r=0}^n \sum_{s=0}^n (C_r + C_s)$

$$\begin{aligned} &= \sum_{r=0}^n \sum_{s=0}^n C_r + \sum_{r=0}^n \sum_{s=0}^n C_s = \sum_{s=0}^n \left(\sum_{r=0}^n C_r \right) + \sum_{r=0}^n \left(\sum_{s=0}^n C_s \right) = \sum_{s=0}^n 2^n + \sum_{r=0}^n 2^n \\ &= \sum_{r=0}^n \sum_{s=0}^n C_r + \sum_{r=0}^n \sum_{s=0}^n C_s = \sum_{s=0}^n \left(\sum_{r=0}^n C_r \right) + \sum_{r=0}^n \left(\sum_{s=0}^n C_s \right) = \sum_{s=0}^n 2^n + \sum_{r=0}^n 2^n \\ &= (n+1) 2^n + (n+1) 2^n = (n+1) 2^{n+1} \end{aligned}$$

Hence (A) is correct answer.

Illustration :

If $(1+x)^n = C_0 + C_1x + \dots + C_nx^n$, then the value of $\sum_{0 \leq r < s \leq n} C_r C_s$ is equal to

- (A) $\frac{1}{2} [2^{2n} - {}^{2n}C_n]$ (B) $\frac{1}{4} [2^{2n} - {}^{2n}C_n]$ (C) $\frac{1}{2} [2^{2n} + {}^{2n}C_n]$ (D) $\frac{1}{2} [2^n - {}^{2n}C_n]$

Sol. We have

$$\sum_{r=0}^n \sum_{s=0}^n C_r C_s = \left(\sum_{r=0}^n C_r^2 \right) + 2 \sum_{0 \leq r < s \leq n} C_r C_s$$

$$\Rightarrow 2^{2n} = {}^{2n}C_n + 2 \sum_{0 \leq r < s \leq n} C_r C_s$$

$$\Rightarrow \sum_{0 \leq r < s \leq n} C_r C_s = \frac{1}{2} [2^{2n} - {}^{2n}C_n]$$

Hence (A) is correct answer.

3.3 AN IMPORTANT CONCEPT :

Finding nature of integral part of expression.

$$N = (a + \sqrt{b})^n \quad (n \in \mathbb{N})$$

Step-1: Consider $N' = (a - \sqrt{b})^n$ or $(\sqrt{b} - a)^n$ according as $a > \sqrt{b}$ or $\sqrt{b} > a$.

Step-2: Use $N + N'$ or $N - N'$ such that result is integer.

Step-3: Use fact $N = I + f$

'I' stands for $[N]$ and 'f' for $\{N\}$.

Illustration :

For $n \in \mathbb{N}$ prove that integral part of $N = (3 + \sqrt{7})^n$ is an odd integer.

Sol. Consider $N' = (3 - \sqrt{7})^n$

$$N = {}^nC_0 3^n + {}^nC_1 3^{n-1} \sqrt{7} + {}^nC_2 3^{n-2} (\sqrt{7})^2 + \dots - {}^nC_n (\sqrt{7})^n.$$

$$N' = {}^nC_0 3^n - {}^nC_1 3^{n-1} \sqrt{7} + {}^nC_2 3^{n-2} (\sqrt{7})^2 + \dots - {}^nC_n (\sqrt{7})^n.$$

$$\text{Using } N + N' = 2 [{}^nC_0 3^n + {}^nC_2 3^{n-2} (\sqrt{7})^2 + \dots]$$

$$\Rightarrow N + N' = 2k \quad (k \in \mathbb{I}) \Rightarrow I + f + N' = 2k \quad \{N = I + f\}$$

$$0 < N' < 1 \Rightarrow 0 < f + N' < 2$$

But $f + N'$ is self an integer $f + N' = 1 \Rightarrow I + 1 = 2k$ or $I = 2k - 1$.

Hence proved.

Illustration :

Show that integral part of $P = (3\sqrt{3} + 5)^{2n+1}$ ($n \in N$) is an even number.

Sol. Consider $P' = (3\sqrt{3} - 5)^{2n+1}$ here $0 < P' < 1$

$$\text{Using } P - P' = 2 \left[{}^{2n+1}C_1 (3\sqrt{3})^{2n} 5' + {}^{2n+1}C_3 (3\sqrt{3})^{2n-2} (5')^3 + \dots \right]$$

$$\Rightarrow I + f - P' = 2k \quad (k \in N) \quad \{P = I + f\}$$

$$-1 < f - P' < 1 \text{ but } f - P' \text{ is an integer} \Rightarrow f - P' = 0 \Rightarrow I = 2k.$$

Illustration :

Let $N = (7 + 4\sqrt{3})^n = I + f$ ($n \in N$), then find the value of $(1 - f) N$.

Sol. Consider $N' = (7 - 4\sqrt{3})^n$

$$\text{Using } N + N' = 2 \left[{}^nC_0 7^n + {}^nC_2 7^{n-2} (4\sqrt{3})^2 + \dots \right]$$

$$\Rightarrow I + f + N' = 2k \quad (k \in N)$$

Illustration :

Let $N = (7 + 4\sqrt{3})^n = I + f$ ($n \in N$), then find the value of $(1 - f) N$.

Sol. Consider $N' = (7 - 4\sqrt{3})^n$

$$\text{Using } N + N' = 2 \left[{}^nC_0 7^n + {}^nC_2 7^{n-2} (4\sqrt{3})^2 + \dots \right]$$

$$\text{or } I + f + N' = 2k \quad (k \in I)$$

$$\text{or } f + N' = 2k - I$$

$$\text{but } 0 < f + N' < 2 \Rightarrow f + N' = 1 \text{ or } f = 1 - N'.$$

$$(1 - f)N = N' N = (7 - 4\sqrt{3})^n (7 + 4\sqrt{3})^n = (49 - 48)^n = 1.$$

Solved Examples

Single correct question

Q.1 Value of middle term of expansion $\left(\frac{2}{3}x^2 - \frac{3}{2x}\right)^{20}$ for $x = p$ is $32 \cdot {}^{20}C_{10}$ value of 'p' is

- (A) 2 (B) $\frac{1}{2}$ (C) $\sqrt{2}$ (D) $\frac{1}{4}$

Sol. Here $n = 20$ so no of terms will be 21 then middle term $t_m = t_{10+1} = {}^{20}C_{10}x^{10}$

Q.2 If $b_1, b_2, \dots, b_n$ are in G.P. with common ratio '2' then $b_1 {}^nC_1 + b_2 {}^nC_2 + \dots + b_n {}^nC_n =$

- (A) $\frac{b_1}{2} (3^n)$ (B) $b_1 (3^n)$ (C) $b_1 (3^n - 1)$ (D) $\frac{b_1}{2} (3^n - 1)$

Sol. $b_1 {}^nC_1 + b_2 {}^nC_2 + \dots + b_n {}^nC_n = b_1 ({}^nC_1 + 2({}^nC_2) + 2^2 ({}^nC_3) + \dots + 2^{n-1} {}^nC_n)$

$$= \frac{b_1}{2} [(1 + 2)^n - 1] = \frac{b_1}{2} (3^n - 1)$$

Q.3 If $(1 + ax)^n = 1 + 8x + 24x^2 + \dots$ then $\frac{a-n}{a+n}$ is equal to

- (A) 3 (B) -3 (C) $-\frac{1}{3}$ (D) $\frac{1}{3}$

Sol. $1 + ax \cdot {}^nC_1 + {}^nC_2 (ax)^2 + \dots = 1 + 8x + 24x^2 + \dots$

By comparison

$$\Rightarrow {}^nC_1 a = 8 \quad \dots\dots(1) \quad \text{and} \quad {}^nC_2 a^2 = 24 \quad \dots\dots(2)$$

By equation (1) and (2), we get

$$n = 4 \text{ and } a = 2.$$

Q.4 In the expansion $\left(x^2 + \frac{2}{x}\right)^n$ ($n \in \mathbb{N}$) has 13th term independent of x, then sum of even divisors of n is

equal to –

- (A) 39 (B) 63 (C) 13 (D) 26

Sol. $t_{13} = {}^nC_{12} \cdot 2^{12} x^{2n-36} \Rightarrow 2n - 36 = 0 \text{ or } n = 18$
 $n = 18 = 2^1 \times 3^2$
 sum of divisors (even) $= (2^1)(3^0 + 3^1 + 3^2)$
 $= 2 \times 13 = 26$

Q.5 Coefficient of x^7 in the expansion of $(1 - x^4)(1 + x)^9$ is –
 (A) 27 (B) -24 (C) 48 (D) -48

Sol. $(1 - x^4)(1 + x)^9 = (1 + x)^9 - x^4(1 + x)^9$
 coefficient of $x^7 = {}^9C_7 - {}^9C_3 = -48$

Q.6 Coefficient of x^{100} in $1 + (1 + x) + (1 + x)^2 + (1 + x)^3 + \dots + (1 + x)^n$ ($n > 100$) is ${}^{201}C_{101}$, then $n =$
 (A) 100 (B) 200 (C) 101 (D) None of these

Sol. Coefficient of $x^{100} = {}^{100}C_{100} + {}^{101}C_{100} + \dots + {}^nC_{100}$
 $= {}^{n+1}C_{101} = {}^{201}C_{101} \Rightarrow n = 200$

Multiple correct type question

Q.7 Coefficients of three consecutive terms in the expansion of $(1 + x)^n$ are in the ratio $1 : 7 : 42$ value of 'n'

Multiple correct type question

Q.7 Coefficients of three consecutive terms in the expansion of $(1 + x)^n$ are in the ratio $1 : 7 : 42$, value of 'n' is always less than or equal to –
 (A) 55 (B) 54 (C) 56 (D) 51

Sol. ${}^nC_{r-1} : {}^nC_r : {}^nC_{r+1} = 1 : 7 : 42 \Rightarrow n = 55 \text{ and } r = 7$

Q.8 $\alpha = \left\{ \frac{3^{200}}{8} \right\}$ where $\{x\}$ fractional of x then possible of middle term in $(2 + 5x)^{72\alpha}$ is –

(A) 5th (B) 7th (C) 8th (D) 6th

Sol. $3^{200} = 9^{100} = (1 + 8)^{100} = 1 + [{}^{100}C_1 8 + {}^{100}C_2 8^2 + \dots]$

$$\Rightarrow \left\{ \frac{3^{200}}{8} \right\} = \frac{1}{8}$$

$$(2 + 5x)^{72\alpha} = (2 + 5x)^9$$

- Q.9 Third term in the expansion of $(x + x^{\log_{10} x})^5$ is 10^6 then possible value of x are –
 (A) 1 (B) 10 (C) $10^{-5/2}$ (D) 10^6

Sol. $T_3 = {}^5C_2 x^3 (x^{\log_{10} x})^2 = 10 \cdot x^{3+2\log_{10} x} = 10^6$
 $\Rightarrow x^{3+2\log_{10} x} = 10^5$ taking log on both sides given

$$(3 + 2 \log_{10} x) \log_{10} x = 5 \Rightarrow \log_{10} x = 1 \quad \text{or} \quad -\frac{5}{2}$$

- Q.10 If $(1 + x + x^2 + x^3) = C_0 + C_1 x + C_2 x^2 + \dots + C_{3n} x^{3n}$ then which of following are correct –
 (A) $C_0 + C_1 + C_2 + \dots + C_{3n} = 2^{2n}$ (B) $C_0 + C_2 + C_4 + \dots = C_1 + C_2 + C_5$
 (C) $C_0 = C_{3n}, C_1 = C_{3n-1}, C_2 = C_{3n-2}$ (D) None of these

Sol. Put $x = 1$ to get $4^n = C_0 + C_1 + C_2 + \dots + C_{3n} \dots$
 Put $x = -1$ to get $\Rightarrow C_0 + C_2 + C_4 + \dots = C_1 + C_3 + C_5 + \dots$

Paragraph type

Paragraph type

Paragraph for question nos. 11 to 13

Let P be a product given by $P = (x + \alpha_1)(x + \alpha_2) \dots (x + \alpha_n)$ and $S_1 = \alpha_1 + \alpha_2 + \dots + \alpha_n = \sum_{i=1}^n \alpha_i$

$$S_2 = \sum_{i < j} \alpha_i \alpha_j \quad S_3 = \sum_{i < j < k} \alpha_i \alpha_j \alpha_k \quad \text{and so on then } P = x^n + S_1 x^{n-1} + S_2 x^{n-2} + \dots + S_n$$

- Q.11 Coefficient of x^8 in $(x+2)^2 (x+3)^3 (x+4)^4$ is –
 (A) 26 (B) 27 (C) 28 (D) 29
- Q.12 Coefficient of x^{203} in $(x-1)(x^2-2)(x^3-3) \dots (x^{20}-20)$ is –
 (A) 11 (B) 12 (C) 13 (D) 15
- Q.13 Coefficient of x^{98} in $(x-1)(x-2) \dots (x-100)$ is
 (A) $1^2 + 2^2 + 3^2 + \dots + 100^2$
 (B) $(1+2+3+\dots+100)^2 - (1^2 + 2^2 + \dots + 100^2)$
 (C) $\frac{1}{2} [(1+2+\dots+100)^2 - (1^2 + 2^2 + \dots + 100^2)]$
 (D) None of these

- Sol. (i) $(x^4 + 4 + 4x)(x^3 + 3C_1 x^2 + \dots)(x^4 + {}^4C_1 x^3 + \dots)$ expand suitably
(ii) $(x-1)(x^2-2)(x^3-3)\dots(x^{20}-20)$

$$= x^{210} \left[\left(1 - \frac{1}{x}\right) \left(1 - \frac{2}{x^2}\right) \dots \left(1 - \frac{20}{x^{20}}\right) \right]$$

- (iii) Coefficient x^{98} in S_2 in product $(x-1)(x-2)\dots(x-100)$

$$= \frac{1}{2} [(1+2+\dots+100)^2 - (1^2+2^2+\dots+100^2)]$$

Match the Column

Q.14

Column-I	Column-II
(A) Coefficient of x^7 and x^8 are equal in the expansion $\left(3 + \frac{x}{2}\right)^n$ value of n is	(P) 9
(B) ${}^nC_2 = 990$ then n is divisible by	(Q) 45
(B) ${}^nC_2 = 990$ then n is divisible by	(Q) 45
(C) Coefficient of x^2 and x^3 are equal in expansion of $\left(3 + \frac{9x}{7}\right)^m$ then m is greater than or equal to	(R) 7
(D) ${}^pC_6 = {}^{p-1}C_5 + 1$ then p is less than	(S) 55
	(T) 15

- Sol. (A) ${}^nC_7 \cdot \frac{3^{n-7}}{2^7} = {}^nC_8 \cdot \frac{3^{n-8}}{2^8} \Rightarrow n = 55$
(B) ${}^nC_2 = 990 \Rightarrow n = 45$
(C) ${}^mC_2 \cdot \frac{3^{m+2}}{7^2} = {}^mC_3 \cdot \frac{3^{m+3}}{7^3} \Rightarrow m = 9$
(D) ${}^pC_6 = {}^{p-1}C_5 + 1$ or ${}^{p-1}C_6 + {}^{p-1}C_5 = {}^{p-1}C_5 + 1$
 $\Rightarrow {}^{p-1}C_6 = 1$ or $p = 7$

Reasoning type question

Q.15 **Statement-1:** ${}^{20}C_0 {}^{15}C_r + {}^{20}C_1 {}^{15}C_{r-1} + \dots + {}^{20}C_r {}^{15}C_0$ has maximum value ${}^{35}C_{18}$.

Statement-2: ${}^nC_0 {}^mC_s + {}^nC_1 {}^mC_{s-1} + {}^nC_2 {}^mC_{s-2} + \dots + {}^nC_s {}^mC_0 = {}^{m+n}C_s$

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

Sol. S_2 is true as $\sum \sum {}^nC_{r_1} {}^mC_{r_2}$ where $r_1 + r_2 = k$ (constant) is equal to ${}^{m+n}C_k$. Sum given in S_1 is equal to ${}^{35}C_{18}$.

Q.16 **Statement-1:** If ${}^{100}C_r, {}^{100}C_{r+1}, {}^{100}C_{r+2}$ and ${}^{100}C_{r+3}$ are in AP then $r = 49$.

Statement-2: Four consecutive binomial coefficient can never be in AP.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.

- (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

Sol. $2 \cdot {}^{100}C_{r+1} = {}^{100}C_r + {}^{100}C_{r+2}$

$$\Rightarrow 2 = \frac{r+1}{100-r} + \frac{99-r}{r+2}$$

which has no solution

Appendix - I

THE PRINCIPLE OF MATHEMATICAL INDUCTION :

Mathematical induction is a technique for proving any statement any theorem, or a formula that is asserted about every natural number.

By "every", or "all" natural numbers, we mean any one that we might possibly name.

For example,

$$1 + 2 + 3 + \dots + n = \frac{1}{2n}(n+1)$$

This asserts that the sum of consecutive numbers from 1 to n is given by the formula on the right. We want to prove that this will be true for $n = 1$, $n = 2$, $n = 3$, and so on. Now we can test the formula for any given number, say $n = 3$.

$$1 + 2 + 3 = \frac{1}{2} \cdot 3 \cdot 4 = 6$$

Which is true. It is also true for $n = 4$.

Which is true. It is also true for $n = 4$.

$$1 + 2 + 3 + 4 = \frac{1}{2} \cdot 4 \cdot 5 = 10.$$

But how are we to prove this rule for every value of n ?

The method of proof is the following. It is called the principle of mathematical induction.

AXIOM OF INDUCTION :

- (1) when a statement is true for a natural number $n = k$,
then it will also be true for its successor, $n = k + 1$,
and
- (2) the statement is true for $n = 1$.
then the statement will be true for every natural number n .

To prove a statement by induction, we must prove parts 1 and 2 above. For, when the statement is true for $n = 1$, then according to 1, it will also be true for 2. But that implies it will be true for 3 which implies it will be true for 4. And so on. It will be true for any natural number that we might name.

Appendix - II

BINOMIAL THEOREM FOR ANY INDEX :

When n is a negative integer or a fraction then the expansion of a binomial is possible only when

- (i) Its first term is 1, and
- (ii) Its second term is numerically less than 1.

Thus when $n \notin \mathbb{N}$ and $|x| < 1$, then it states

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots + \frac{n(n-1)(n-r+1)}{r!}x^r + \dots \infty$$

1.1 General Term :

$$T_{r+1} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} \cdot x^r$$

Note :

- (i) In this expansion the coefficient of different terms can not be expressed as ${}^nC_0, {}^nC_1, {}^nC_2, \dots$ because n is not a positive integer.
- (ii) In this case there are infinite terms in the expansion.

- (ii) In this case there are infinite terms in the expansion.

1.2 Some Important Expansions :

If $|x| < 1$ and $n \in \mathbb{Q}$ but $n \notin \mathbb{N}$, then

(a) $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots + \frac{n(n-1)\dots(n-r+1)}{r!}x^r + \dots$

(b) $(1-x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + \dots + \frac{n(n-1)\dots(n-r+1)}{r!}(-x)^r + \dots$

(c) $(1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \dots + \frac{n(n+1)\dots(n+r-1)}{r!}x^r + \dots$

(d) $(1+x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \dots + \frac{n(n+1)\dots(n+r-1)}{r!}(-x)^r + \dots$

By putting $n = 1, 2, 3$ in the above results (c) and (d), we get the following results-

(e) $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots + x^r + \dots$

General term $T_{r+1} = x^r$

(f) $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + (-x)^r + \dots$

General term $T_{r+1} = (-x)^r$

- (g) $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots + (r+1)x^r + \dots$
General term $T_{r+1} = (r+1)x^r$
- (h) $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots + (r+1)(-x)^r + \dots$
General term $T_{r+1} = (r+1)(-x)^r$
- (i) $(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \dots + \frac{(r+1)(r+2)}{2!}x^r + \dots$
General term $= \frac{(r+1)(r+2)}{2!}x^r$
- (j) $(1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \dots + \frac{(r+1)(r+2)}{2!}(-x)^r + \dots$
General term $= \frac{(r+1)(r+2)}{2!}(-x)^r$

Illustration :

If $|x| < 2/3$ then the fourth term in the expansion of $\left(1 + \frac{3}{2}x\right)^{1/2}$ is—

- (A) $\frac{27}{128}x^3$ (B) $-\frac{27}{128}x^3$ (C) $\frac{81}{256}x^3$ (D) $-\frac{81}{256}x^3$

Sol. $T_4 = \frac{1/2(1/2-1)(1/2-2)}{3!} \cdot \left(\frac{3x}{2}\right)^3 = \frac{27}{128}x^3$ **Ans. [A]**

- (A) $\frac{27}{128}x^3$ (B) $-\frac{27}{128}x^3$ (C) $\frac{81}{256}x^3$ (D) $-\frac{81}{256}x^3$

Sol. $T_4 = \frac{1/2(1/2-1)(1/2-2)}{3!} \cdot \left(\frac{3x}{2}\right)^3 = \frac{27}{128}x^3$ **Ans. [A]**

Illustration :

The term independent of x in the expansion of $\left(\frac{1-x}{1+x}\right)^2$ is—

- (A) 4 (B) 3 (C) 2 (D) 1

Sol. $(1-x)^2(1+x)^{-2}$
 $\Rightarrow (1-2x+x^2)(1-2x+x^2+\dots)$
 $\Rightarrow$ so term independent of $x = 1$. **Ans. [D]**

Illustration :

The coefficient of x^5 in the expansion of $(1-x)^{-6}$ is—

- (A) 1260 (B) -1260 (C) -252 (D) 252

Sol. x^5 occurs in T_6 of the expansion, so

$$T_6 = T_{5+1} = \frac{6 \cdot 7 \cdot 8 \cdot 9 \cdot 10}{5!} x^5 = 252 x^5$$

$\therefore$ Coefficient of $x^5 = 252$ **Ans. [D]**

1.3 Applications of binomial theorem :

- (I) With the help of binomial theorem, we can find out the value of sq. root, cube root and 4th root etc. of the given number upto any decimal places.

Illustration :

The value of cube root of 1001 upto five decimal places is –

- (A) 10.03333 (B) 10.00333 (C) 10.00033 (D) None of these

Sol. $(1001)^{1/3} = (1000+1)^{1/3} = 10 \left(1 + \frac{1}{1000} \right)^{1/3} = 10 \left\{ 1 + \frac{1}{3} \cdot \frac{1}{1000} + \frac{1/3(1/3-1)}{2!} \frac{1}{1000^2} + \dots \right\}$

$$= 10 \{ 1 + 0.0003333 - 0.00000011 + \dots \}$$

$$= 10.00333 \quad \text{Ans. [B]}$$

- (II) To find the sum of Infinite series :

We can compare the given infinite series with the expansion of $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$

We can compare the given infinite series with the expansion of $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$

$x^2 + \dots$ and by finding the value of x and n and putting in $(1+x)^n$ the sum of series is determined.

Illustration :

The sum of $1 + \frac{1}{4} + \frac{1.3}{4.8} + \frac{1.3.5}{4.8.12} + \dots \infty$ is –

- (A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\sqrt{3}$ (D) $2^{3/2}$

Sol. Comparing with $1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$

$$nx = 1/4 \quad \dots(1)$$

and $\frac{n(n-1)x^2}{2!} = 1.3/4.8$

or $\frac{nx(nx-x)}{2!} = \frac{3}{32} \Rightarrow \frac{1}{4} \left(\frac{1}{4} - x \right) = \frac{3}{16}$ (by (1))

$\Rightarrow \left(\frac{1}{4} - x \right) = \frac{3}{4} \Rightarrow x = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$... (2)

putting the value of x in (1)

$n(-1/2) = 1/4 \Rightarrow n = -1/2$

$\therefore$ sum of series $= (1+x)^n$
 $= (1-1/2)^{-1/2} = (1/2)^{-1/2} = \sqrt{2}$

Ans.[A]

(III) Approximation:

Illustration :

If x is so small so that its square and higher power can be neglected.

If x is so small so that its square and higher power can be neglected.

Find the value of $\frac{\left(1 + \frac{2x}{3}\right)^{-5} + (4+2x)^{1/2}}{(4+x)^{3/2}}$.

Sol.
$$\frac{\left(1 + \frac{2x}{3}\right)^{-5} + (4+2x)^{1/2}}{(4+x)^{3/2}} = \frac{\left(1 - \frac{10x}{3}\right) + 2\left(1 + \frac{x}{4}\right)}{8\left(1 + \frac{3x}{8}\right)} = \frac{1}{8} \left(3 - \frac{10x}{3} + \frac{x}{2}\right) \left(1 + \frac{3x}{8}\right)^{-1}$$

$$= \frac{3}{8} \left(1 - \frac{17x}{18}\right) \left(1 - \frac{3x}{8}\right) = \frac{3}{8} \left(1 - \frac{17x}{18} - \frac{3x}{8}\right) = \frac{72-95x}{24 \times 8}$$